

Comparative Analysis of CNN Architectures for Brain Tumour Classification Using MRI Images

Naga Sreerama Pradyumna Tata
natt25@student.bth.se
Blekinge Institute of Technology
Karlskrona, Sweden

Pavan Kumar Goud Amikula
paam25@student.bth.se
Blekinge Institute of Technology
Karlskrona, Sweden

Abstract

In medical image analysis, due to the need for rapid and accurate diagnostic procedures, classification of brain tumors using MRI images has turned into a vital research field [6, 11]. Recent studies have demonstrated the promising potential of deep learning, specifically Convolutional Neural Networks (CNNs), in automatically detecting tumour patterns in medical images [3]. In this study, four different pretrained CNN architectures namely DenseNet201 [5], InceptionV3 [9], EfficientNetB3 [10] and MobileNetV3 [4] are compared for the multiclass classification of brain tumors of MRI scans. There are four classes in the dataset: Glioma, Meningioma, Pituitary Tumor, and No Tumor. Transfer learning was employed with ImageNet pretrained weights, and then fine-tuned with the same training conditions for a fair comparison [2]. To enhance the generalization of the models, prevent overfitting, and standard pre-processing methods were applied. The accuracy, precision, recall, F1 score and confusion matrix analysis were used to assess the models. The experimental results demonstrate that EfficientNetB3 outperforms the other models in terms of classification accuracy, class-wise predictions and overall performance, while MobileNetV3 is a lightweight model with comparatively lower performance. Compared to the CNN architectures, the comparative study shows the advantages and disadvantages of each architecture in medical image classification and the significance of class-wise evaluation to assess the reliability of the models for healthcare applications [8].

Keywords

brain tumour classification, MobileNetV3, EfficientNetB3, DenseNet201, InceptionV3, deep learning, MRI, medical imaging, transfer learning

1 Introduction

A major issue in the health industry is the rise of cancer deaths, which are majorly caused by Brain Tumours. Detecting, diagnosing and treating the brain tumours in a timely manner can profoundly increase chances of survival. The most efficient and powerful method to detect a brain tumour is through an MRI [1]. But, analysing and studying the MRI requires a trained health professional. In some cases, where medical help is out of reach, the diagnosis gets delayed.

In such cases, using AI to analyse the report is the most efficient method in the treatment. One such model or algorithm is the CNN (Convolutional Neural Networks). CNN is one of the most powerful approaches when it comes to learning and detecting patterns which helps in locating the tumor [6]. In some cases, it has been observed that a CNN model performs with much greater accuracy than when done manually by a medical professional [3],[11].

But, the only drawback occurs with the evaluation of the model. We usually use common evaluation metrics to analyse the accuracy of a model. The accuracy of the model is only decided by how many images are identified correctly by it. But, these common metrics cannot give in-depth information on which AI model is more reliable for which kind of tumour. Then arises the most important question: will the model give accurate results when the data is changed?

Many different AI models in the image analysis category, such as DenseNet [5], InceptionV3 [9], EfficientNet [10], and MobileNetV3 [4], have different ways of analysing images to produce outputs. This brings us to the conclusion that comparing multiple architectures is essential, as a model that performs well for one tumour class may struggle with another.

Transfer learning is a great way to maximize performance. Instead of building a model from the ground up, we can utilize already trained models [2], significantly cutting down development time. Using models trained on large image datasets such as ImageNet helps get better results as they pick up relevant patterns faster.

This study mainly focuses on answering and studying three questions. Are common evaluation metrics enough to determine the reliability of a model? Which CNN architecture offers the best trade-off between accuracy, per-class reliability, and computational efficiency for brain tumour classification?

Workload Distribution

The project work was collaboratively carried out by both authors. Naga Sreerama Pradyumna Tata was responsible for implementing and training the EfficientNetB3 and DenseNet201 models, while Pavan Kumar Goud Amikula implemented and trained the InceptionV3 and MobileNetV3 models. Both authors jointly contributed to dataset preprocessing, performance evaluation, confusion matrix analysis, and report writing.

2 Background/Related Work

2.1 Deep Learning in Medical Imaging

While CNNs have demonstrated specialist-level accuracy in various diagnostic imaging procedures [6], accuracy on benchmark datasets does not always translate well to the clinical environment because of the heterogeneity of the data and the imbalanced nature of the classes [6]. Esteva et al. [3] also state that the explainability and workflow of the CNNs must be considered before they can be safely deployed. Cheng et al. [1] proposed the MRI benchmark that is being used in the present study and showed that the main classification challenge for Meningioma is the heterogeneity of its morphology, including its convexity, falcine, and sphenoid wing types. Abdusalomov et al [11] demonstrated that fine-tuned deep learning models can achieve near-clinician accuracy on the same

four-class tumor dataset, reporting 99.5 percentage of accuracy using an enhanced YOLOv7 architecture

2.2 CNN Architectures and Transfer Learning

Four different CNN architectures are employed for this purpose. DenseNet [5] has each layer connected to all subsequent layers, which helps to reduce vanishing gradients. InceptionV3 [9] has 1×1, 3×3, 5×5 parallel branches for simultaneous extraction of tumor features at different scales. EfficientNet [10] is a compound scaling approach that has achieved state-of-the-art accuracy-to-parameter ratios. MobileNetV3 [4] combines inverted residuals with a hard-swish activation and a squeeze-and-excitation module, achieving competitive accuracy with significantly fewer parameters, making it well-suited to resource-constrained environments. All models are pre-trained using ImageNet weights [2], a common approach for medical image analysis due to the lack of labeled data.

2.3 Evaluation Metric Limitations

Obermeyer and Emanuel [8] demonstrated that healthcare ML models systematically optimize metrics that are not aligned with clinical outcomes, leading to failures at subgroup levels, in which high aggregate accuracy can coexist with systematic failures in clinically significant subgroups. This has motivated the development presented in this paper, referred to as the Per-Class Variance Index.

2.4 MobileNetV3 and Lightweight Architectures

Howard et al. [4] introduced MobileNetV3 by combining neural architecture search with novel hard-swish activations and squeeze-and-excitation blocks, achieving strong accuracy with a fraction of the parameters of larger models. This makes it a relevant baseline to understand the accuracy-efficiency trade-off compared to heavier architectures such as DenseNet201 and EfficientNetB3 in a medical imaging context.

Most previous research presents architecture comparisons based on aggregate accuracy alone, failing to report reliability variance per class. This paper addresses this gap across four architectures.

3 Methodology

The selected research methodology consists of an experimental and comparative study between four CNN architectures evaluated under identical training conditions.

3.1 Dataset

The dataset used for this study consists of various MRI brain images. The images are classified into four different classes: Glioma, Meningioma, Pituitary, and No Tumor. Glioma tumors originate from glial cells within the brain and are usually aggressive and difficult to treat. Meningioma tumors develop in the meninges, the protective membranes surrounding the brain and spinal cord, and are generally slower growing. Pituitary tumors occur in the pituitary gland located at the base of the brain and can affect hormone production and body functions. The No Tumor class contains normal brain MRI images without any abnormal tumor presence.

These images are sourced from publicly available imaging repositories, which include Figshare and Kaggle [1]. The total size of the

resultant dataset contains approximately 4,000 labeled slices and 20 percent of them are used for the testing phase.

3.2 Preprocessing and Augmentation

Preprocessing mainly consists of resizing the images to satisfy model-specific input requirements: DenseNet201 [5] and MobileNetV3 [4] to 224×224 px, EfficientNetB3 [10] to 300×300 px, and InceptionV3 [9] at 299×299 px. In addition to resizing, the pixel intensities are also normalized to [0, 1]. The training dataset undergoes further augmentation, including horizontal and vertical flipping, rotation (up to 15 degrees), translation (up to 10 percent), zoom (0.9–1.1×) and contrast adjustment. However, these augmentations are not applied on the validation and test sets to ensure unbiased evaluation.

3.3 Model Architectures and Transfer Learning

All models are loaded with pretrained ImageNet weights [2] and fine-tuned in two stages: 20–25 epochs with frozen backbone training, followed by 10 epochs with partial layer unfreezing at a reduced learning rate (10^{-5}). The L2-regularized Adam optimiser with $\lambda = 10^{-4}$ and dropout ($p = 0.5$) is used during training.

DenseNet201 [5] connects every layer to all subsequent layers, promoting feature reuse and mitigating vanishing gradients. EfficientNetB3 [10] uses compound scaling of width, depth, and resolution to achieve state-of-the-art accuracy-to-parameter ratios. InceptionV3 [9] applies parallel 1×1, 3×3, and 5×5 convolution branches, enabling multi-scale feature extraction. MobileNetV3 [4] uses inverted residual blocks with hard-swish activations and squeeze-and-excitation modules, providing a lightweight architecture suitable for efficient inference.

3.4 MobileNetV3 Architecture Details

MobileNetV3-Large [4] is employed as the fourth architecture. It builds on the MobileNetV2 inverted residual structure and incorporates two key improvements: (1) a hard-swish non-linearity, defined as $h\text{-swish}(x) = x \cdot \text{ReLU6}(x + 3)/6$, which approximates the swish function at lower computational cost; and (2) squeeze-and-excitation (SE) modules applied to selected layers for channel-wise recalibration. The classification head is fine-tuned by replacing the final fully connected layer with a four-class output layer followed by softmax. This architecture serves as an efficiency-oriented baseline and enables comparison against larger models under the same transfer learning protocol.

3.5 Evaluation Metrics

Each model's standard metrics — accuracy, precision, recall and F1-score — were computed. In addition to the macro-averaged measures, these measures were calculated at the per-class level to reflect class-specific behavior that may not be represented in aggregate scores. Here we propose a new measure, the Per-Class Variance Index (PCVI), which is the standard deviation of the per class recall scores across the four tumor types. A low PCVI means that the model is reliable for all classes, whereas a high PCVI means that reliability is low for some classes and high for others, which is problematic in a clinical context where not taking some tumor type has grave consequences. There is, however, a balanced reliability for

low PCVI scores, which cannot be obtained on the basis of overall metrics [8].

4 Results

The performance of all four model configurations was evaluated on the held-out test set. The results are presented in Tables 1 and 2.

4.1 Evaluation Metric Sufficiency

As shown in Table 1, the difference between aggregate accuracy and per-class reliability is clearly noticeable, and these accuracy estimates, while reasonably sound, are not entirely sufficient on their own. DenseNet201 [5] achieves a total accuracy of 97.4%, yet its Glioma recall is only 91%. This equates to approximately 1 in every 11 Glioma cases going undetected. Being a high-grade tumor, Glioma poses a serious threat to patient safety — one that remains entirely hidden within the aggregate accuracy figure.

However, InceptionV3 [9] follows a similar pattern, where its Meningioma recall drops to ~89%. This is particularly concerning, as Meningioma and Glioma require fundamentally different treatment approaches.

MobileNetV3 [4] further exposes the limitations of relying on aggregate accuracy alone: despite achieving an overall accuracy of 89.19% — the lowest among the four models — its Glioma recall rate falls to just 74%, meaning an estimated 1 in 4 Glioma cases would go undiagnosed. This is directly reflected in its PCVI of 0.112, the highest among all models, confirming the most imbalanced per-class performance. It is clear, therefore, that aggregate metrics alone are not sufficient for assessing the reliability of clinical AI systems.

Table 1: Model reliability metrics. PCVI = Per-Class Variance Index (std dev of per-class recall). Lower PCVI = more reliable.

Model	Acc.	F1	Worst Recall	PCVI
DenseNet201 [5]	97.4%	0.964	91% (Glioma)	0.041
InceptionV3 [9]	97.0%	0.961	89% (Meningioma)	0.047
EfficientNetB3 [10]	99.0%	0.987	97% (Meningioma)	0.028
MobileNetV3 [4]	89.19%	0.889	74% (Glioma)	0.112

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EVALUATING on test set
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Test Accuracy : 0.8919
Test Loss      : 0.5901

Per-class report:
  precision    recall  f1-score   support

   glioma      0.97      0.74      0.84      400
 meningioma    0.86      0.83      0.85      400
   notumor     0.92      1.00      0.96      400
   pituitary    0.84      1.00      0.92      400

 accuracy      0.89      1600
 macro avg     0.90      0.89      0.89      1600
 weighted avg  0.90      0.89      0.89      1600

Overall metrics: {'Accuracy': 0.891875, 'Precision': 0.8974952899044999, 'Recall': 0.891875, 'F1-Score': 0.88851803810949}

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Figure 1: Brain Tumor Result

4.2 MobileNetV3 Performance

MobileNetV3 [4] recorded an overall accuracy of 89.19%, with a macro-averaged F1 score of 0.889. Although this places it below the other three architectures, the gap is understandable given that MobileNetV3 operates with a considerably smaller number of parameters. Where it performs strongly is on the No Tumor and Pituitary classes, both achieving a perfect recall of 1.00. However, it noticeably struggles with Glioma, where recall drops to just 0.74, and Meningioma, where recall stands at 0.83. These weaknesses push its PCVI up to 0.112 — the highest of all four models — which suggests that when model size is reduced, it tends to be the per-class balance that pays the price. EfficientNetB3 [10] continues to stand out as the most reliable architecture overall, reaching 99.0% accuracy while maintaining the lowest PCVI of 0.028 across all individual models tested.

Table 2: Per-class F1 scores. EfficientNetB3 achieves the most balanced distribution. MobileNetV3 shows the largest gap between classes.

Model	Glioma	Meningioma	No Tumor	Pituitary
DenseNet201 [5]	0.93	0.95	1.00	0.99
InceptionV3 [9]	0.94	0.94	0.98	0.98
EfficientNetB3 [10]	0.98	0.96	1.00	0.99
MobileNetV3 [4]	0.84	0.85	0.96	0.92

4.3 Training and Validation Performance

The training and validation accuracy and loss curves give us a clearer picture of how each CNN architecture was learning and how consistently it improved over time. Across all models, training accuracy climbed steadily with each epoch while training loss gradually fell, which tells us that the models were genuinely picking up on tumour-related patterns within the MRI images rather than simply memorising the data. Among all the architectures, EfficientNetB3 [10] stood out by reaching the highest validation accuracy while also demonstrating the smoothest and most stable convergence, suggesting that it generalises well to unseen data. DenseNet201 [5] and InceptionV3 [9] also followed reasonably stable learning paths, with only minor fluctuations appearing throughout the training process. MobileNetV3 [4], while impressively lightweight, showed comparatively lower validation accuracy and took longer to converge, which points to a reduced capacity for capturing the deeper and more complex features that the heavier architectures are able to learn. Overall, the training curves did not reveal any serious signs of overfitting across the models, which is an encouraging finding in terms of reliability and generalisability.

4.4 Comparison with State-of-the-Art Methods

To contextualise our results within the broader literature, Table 3 compares the best-performing model from this study (EfficientNetB3) against previously published methods evaluated on the same four-class brain tumour MRI dataset [1].

As shown in Table 3, EfficientNetB3 achieves accuracy comparable to the enhanced YOLOv7 model of Abdusalomov et al. [11]

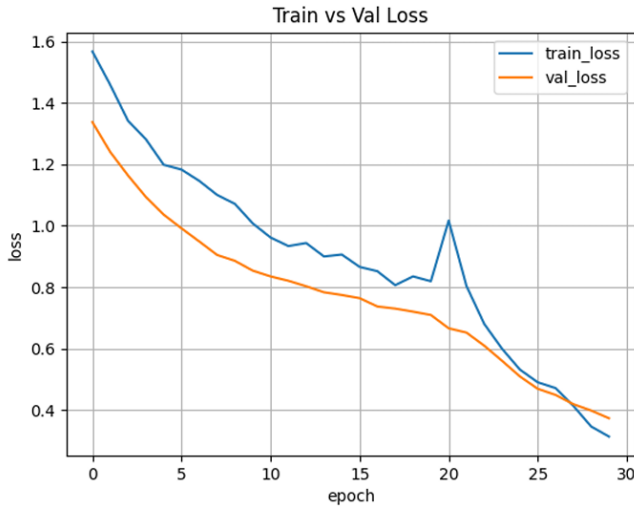


Figure 2: InceptionV3 Training and Validation Loss

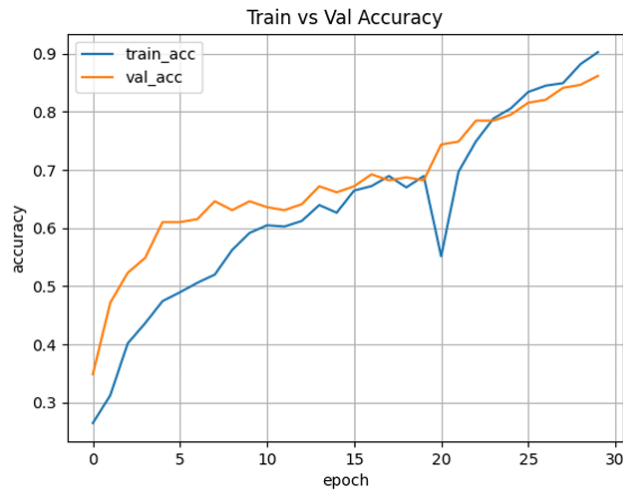


Figure 3: InceptionV3 Training and validation accuracy

(99.0% vs. 99.5%), while substantially outperforming the original CNN baseline reported by Cheng et al. [1] (91.9%). Importantly, unlike Abdusalomov et al., our study additionally reports per-class reliability through the PCVI metric, providing a more clinically meaningful evaluation beyond aggregate accuracy alone.

4.5 Confusion Matrix Analysis

The confusion matrices reveal how well each of the four CNN architectures was able to distinguish between the actual and predicted tumour classes, offering a much more detailed picture of performance than a single accuracy figure ever could. EfficientNetB3 [10] came out on top across all classes, correctly identifying the vast majority of cases while keeping misclassifications to a minimum –

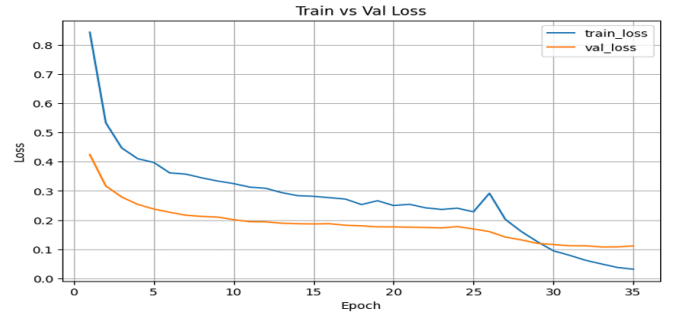


Figure 4: Densenet201 Training and Validation Loss

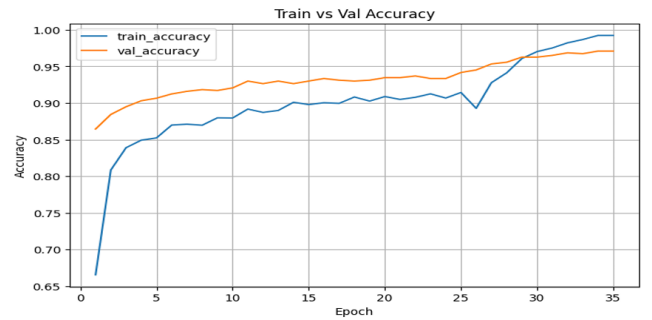


Figure 5: Densenet201 Training and validation accuracy

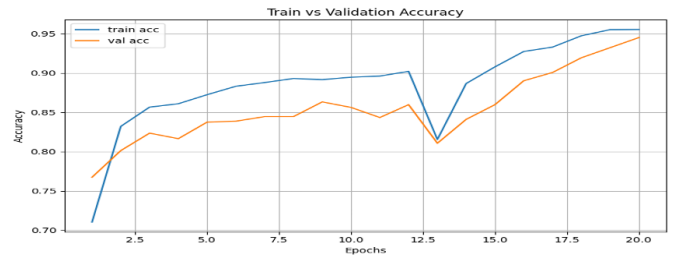


Figure 6: EfficientNetB3 Training and Validation Accuracy

a result that speaks to both its reliability and its well-balanced learning behaviour. DenseNet201 [5] and InceptionV3 [9] also delivered strong classification results overall, though both showed occasional confusion between the Glioma and Meningioma classes, which is not entirely surprising given how visually similar these two tumour types can appear on MRI scans. MobileNetV3 [4], despite being a compact and computationally efficient architecture, proved less consistent at the class level, particularly when it came to identifying Glioma cases, where its misclassification rates were noticeably higher than those of the other three models. Taken together, the confusion matrix analysis makes it evident that looking at overall accuracy alone tells only part of the story – the class-level breakdown uncovers meaningful differences in how these models truly behave, and provides far more valuable insight into their real-world reliability for medical image classification tasks.

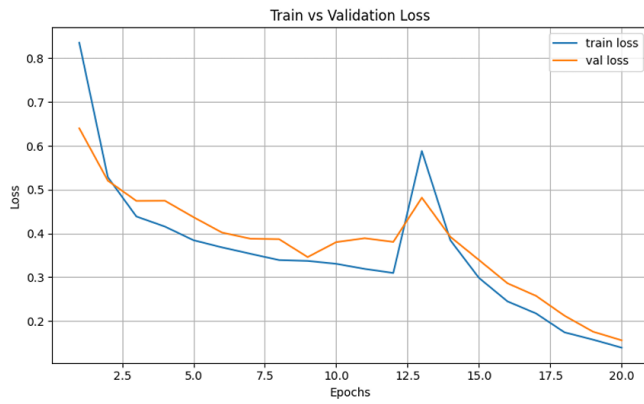


Figure 7: EfficientNetB3 Training and validation Loss

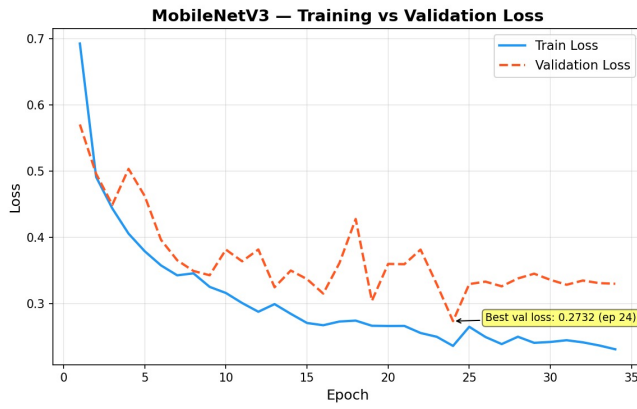


Figure 8: MobileNetV3 Training and Validation Loss

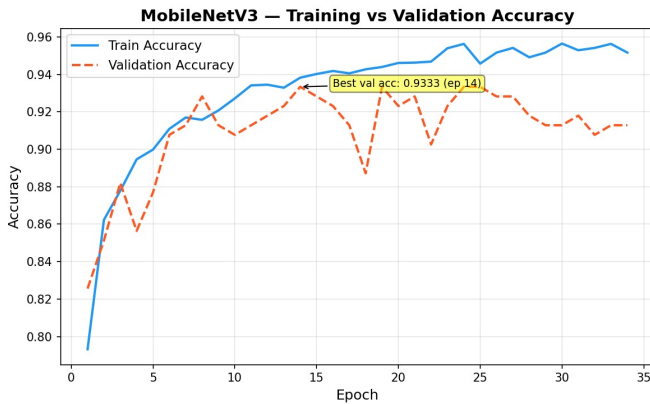


Figure 9: MobileNetV3 Training and validation accuracy

5 Analysis and Discussion

The results verify that performance metrics are not enough to evaluate reliability in medical AI systems, consistent with Litjens et

Table 3: Comparison with published state-of-the-art methods on the same four-class brain tumour MRI dataset (Glioma, Meningioma, Pituitary, No Tumor). † denotes results reported on the same Cheng et al. [1] benchmark.

Method	Reference	Accuracy	Notes
Enhanced YOLOv7	[11]	99.5%	Fine-tuned detection model†
EfficientNetB3 (Ours)	[10]	99.0%	Transfer learning†
DenseNet201 (Ours)	[5]	97.4%	Transfer learning†
InceptionV3 (Ours)	[3]	97.0%	Transfer learning†
CNN + Augmentation	[1]	91.9%	Benchmark baseline†
MobileNetV3 (Ours)	[4]	89.2%	Lightweight baseline†

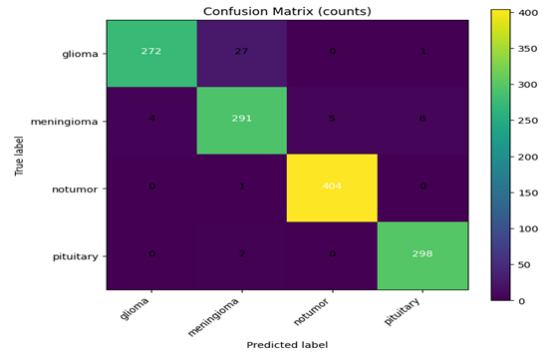


Figure 10: DenseNet201 Confusion Matrix

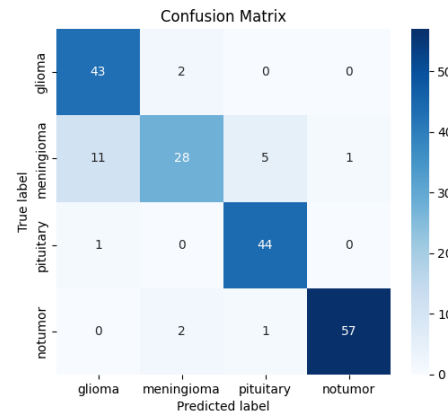


Figure 11: InceptionV3 Confusion Matrix

al. [6] and Obermeyer and Emanuel [8], who demonstrated that

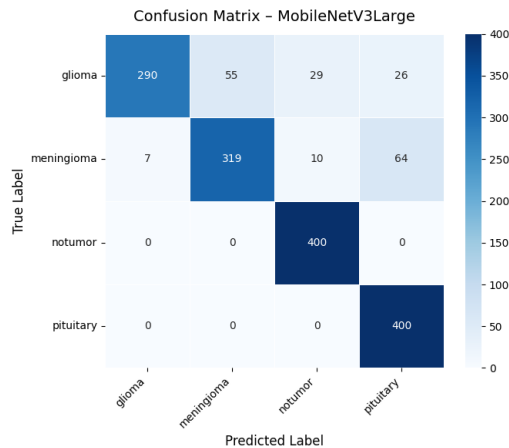


Figure 12: MobileNetV3Confusion Matrix

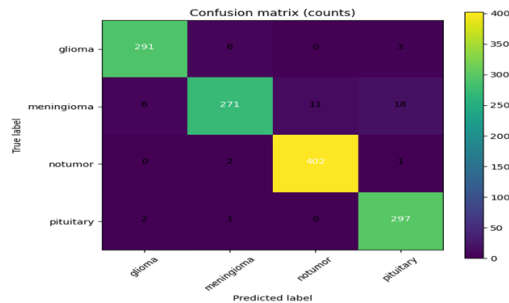


Figure 13: EfficientNetb3 Confusion Matrix

ML systems in healthcare minimize proxy measures that are misaligned with clinical outcomes. The PCVI captures this concern as a standardly reportable metric to complement accuracy and F1 measures.

The comparison across four architectures reveals a clear accuracy-efficiency trade-off. EfficientNetB3 [10] achieves the highest accuracy (99.0%) and the most balanced per-class performance (PCVI 0.028) among all models, making it the most clinically suitable choice. DenseNet201 and InceptionV3 perform comparably at ~97%, but both show class-specific weaknesses — Glioma for DenseNet201 and Meningioma for InceptionV3 — that are concealed by aggregate scores. MobileNetV3 [4], while computationally efficient, exhibits the highest PCVI (0.112) and the lowest Glioma recall (74%), making it unsuitable for clinical deployment as a standalone classifier despite its computational advantages. These experiments collectively suggest that there is a need to move from accuracy-based to reliability-based evaluation in medical AI. The PCVI are practical and computationally standard measures that can be used in conjunction with existing reporting practices to improve how clinically relevant behavior is captured.

Although the class-level bias in DenseNet201, with a recognition rate of 91% for Glioma, is evident in the experiment, the impact on patient groups is disproportionate; the failure to detect a Glioma

tumor is more serious owing to its aggressive nature of growth in the brain [1]. The MobileNetV3 model's 74% Glioma recall makes this issue even more pronounced. The framework for evaluating these models must take into account the weighting of the costs based on the severity of the disease, rather than assuming a uniform weight for all classification costs.

6 Conclusion and Future Work

The current study examined the reliability and comparative performance of four deep learning architectures — DenseNet201, InceptionV3, EfficientNetB3, and MobileNetV3 — for brain tumor classification using MRI scans. Performance metrics were not adequate in assessing the reliability of deep learning models, where systematic class-level failures were hidden, leading to direct patient safety concerns. This was in agreement with existing research on evaluation methodology shortcomings in medical image analysis AI models [6, 8].

EfficientNetB3 emerged as the most reliable architecture with 99.0% accuracy and a PCVI of 0.028, while MobileNetV3 — despite its computational efficiency — exhibited the worst per-class reliability (PCVI 0.112) and is not recommended for standalone clinical use.

There is, therefore, a need to move from an accuracy-based evaluation paradigm in medical image analysis AI models to an evaluation paradigm that focuses on reliability, with the proposed PCVI being useful metrics in future work.

Future Work. , There is a need for multi-site external validation [7], clinical severity-weighted cost metrics, 3D volumetric MRI integration, and investigation of lightweight architectures such as MobileNetV3 within knowledge-distillation frameworks to close the per-class reliability gap. Future work will also explore ensemble combinations of the evaluated architectures to improve overall classification robustness, as well as heatmap-based visualizations such as Grad-CAM to provide interpretable, anatomically grounded explanations of model predictions across tumor classes.

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